

Donggeon Bae

Integrated M.S./Ph.D. Student · Mechanical Engineering
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RESEARCH INTERESTS

Computational imaging · Computer vision · Physics-guided deep learning · Lensless reconstruction · Biomedical imaging

EDUCATION

Seoul National University · 2025-present
Integrated M.S./Ph.D., Mechanical Engineering
Adviser: Prof. Seung Ah Lee · Entered Fall 2025

Yonsei University · 2018-2024
B.S., Electrical and Electronic Engineering

RESEARCH EXPERIENCE

Undergraduate Researcher · 2020-2024
Optical Imaging System Laboratory, Yonsei University
Adviser: Prof. Seung Ah Lee

PUBLICATIONS

- 2026
Published

Integrated Forward-Inverse Network for Lensless Image Reconstruction ↗
Donggeon Bae, Jaewoo Jung, Yong Guk Kang, Kyung Lee, Taeyoung Kim, Jongho Kim, Sangjun Byun, Joonsik Park, Seung Ah Lee
Computer Vision - ECCV 2026, Part XVIII · LNCS 17018, pp. 628-645 · Springer
[Preprint ↗](#) [Project ↗](#) [Code ↗](#)
- 2026
Accepted

Physics-guided neural network for single-shot real-time dynamic speckle imaging ↗
Sangjun Byun, Changyoon Yi, **Donggeon Bae**, Yong Guk Kang, Jongho Kim, Hojin Chang, Seung Ah Lee
Journal of Physics: Photonics
- 2026
Published

Geometry-aware phase compensation for sampling-efficient angular spectrum method ↗
Yong Guk Kang, **Donggeon Bae**, Jaewoo Jung, Sangjun Byun, Hyo Jun Ahn, Hojin Chang, Joonsik Park, Hyeonyong Lee, Seung Ah Lee
Optics Express 34(8), 15244-15260
- 2020
Published

Lensless Imaging with an End-to-End Deep Neural Network ↗
Donggeon Bae, Jaewoo Jung, Nakkyu Baek, Seung Ah Lee
IEEE ICCE-Asia, pp. 1-5

TEACHING & SERVICE

Teaching Assistant, Engineering Research Practice I & II (UROP) · Spring 2026

Seoul National University

Coordinated course orientations and recorded materials; guided report submissions and supported course administration.

CONFERENCE PRESENTATIONS

Sep 2026 **Integrated Forward-Inverse Network for Lensless Image Reconstruction** ↗
Poster ECCV 2026 · Malmö, Sweden

2026-01-17 **Interleaved Forward-Inverse Modeling in a Physics-Guided Neural Network for Computational Image Reconstruction**
Oral SPIE Photonics West · San Francisco · 13865-26

2025-10-31 **Recurrent Attentive Reconstruction Network for Progressive Lensless Imaging**
Poster SPIE ABC · Jeju

2024-11-01 **DiffLIR: Deep Deconvolution Diffusion Network for Lensless Image Reconstruction**
Poster ABC · Gyeongju

2024-08-07 **D3Net: Lensless Reconstruction via Deep Deconvolution Diffusion Network**
Poster CLEO-PR · Incheon

2023-08-29 **Learned Spatially Varying Forward Model for Lensless Imaging**
Poster OPC · Seogwipo

2020-11-02 **Lensless Imaging with an End-to-End Deep Neural Network**
Oral ICCE-Asia · Busan

AWARDS

August 2023 **Best Poster Award**
Award Optics and Photonics Congress · Optical Society of Korea

Updated 2026-10-01. Accepted and published works are labeled separately. Public profile: imaging.snu.ac.kr/team.